UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

CS-305L: Parallel & Distributed Computing Lab

SUBMITTED BY

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SECTION : CS-A

SEMESTER : 6th

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LAB 5

Task 1

What is the role of MPI_Barrier in collective communication?

Answer:

MPI_Barrier is a synchronization function that forces all processes to stop at a certain point until every process reaches that point. Only after all processes arrive at the barrier, they continue execution. This ensures coordinated execution and prevents some processes from running ahead of others.

Task 2

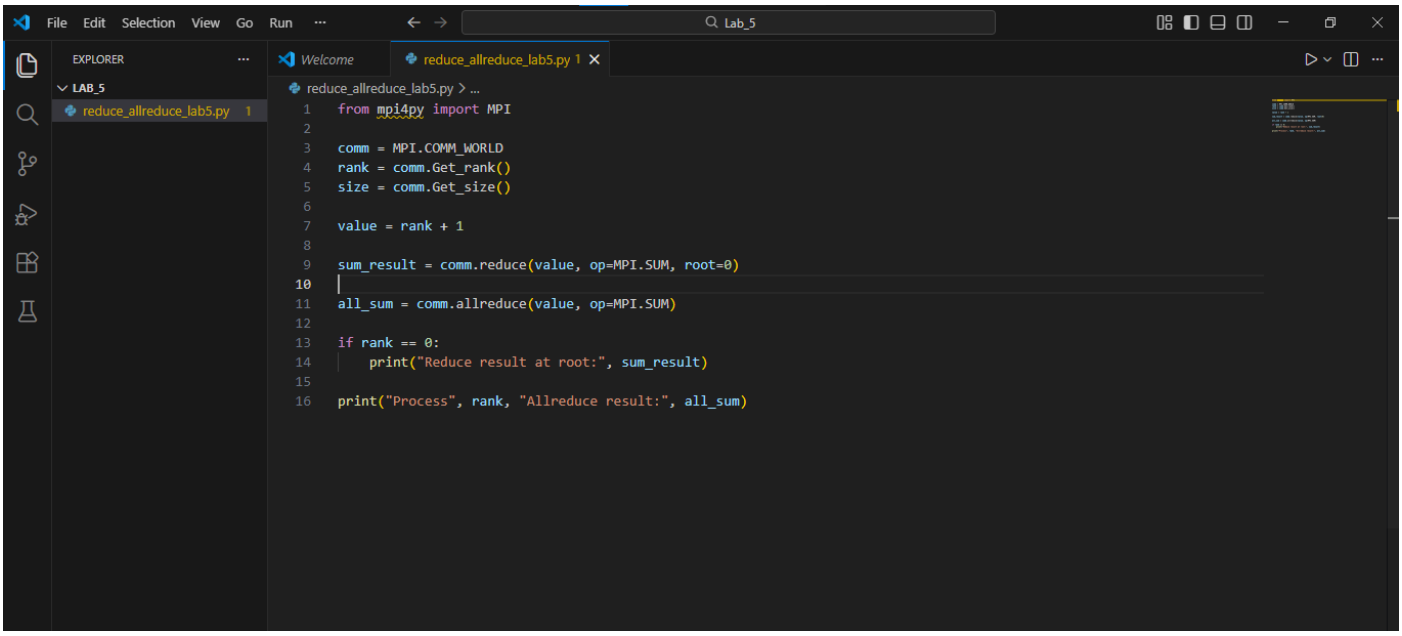
Why is synchronization important in parallel programming?

Answer:

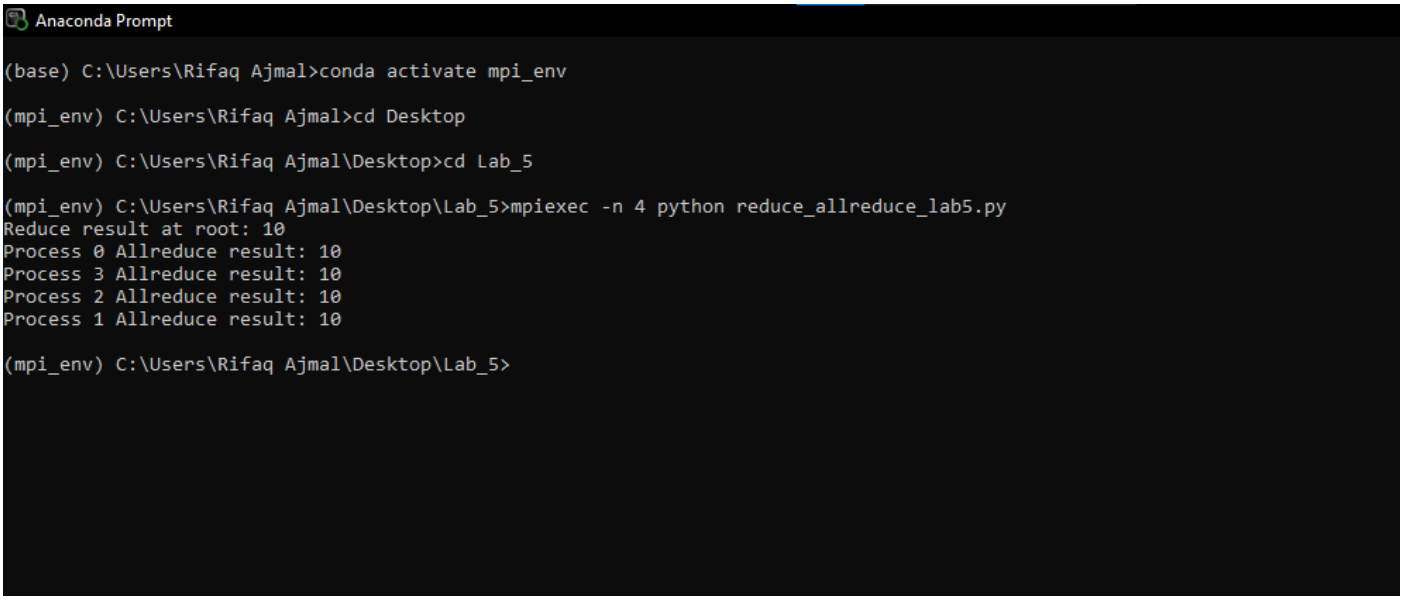
Synchronization is important in parallel programming because multiple processes run at the same time. Without synchronization, processes may access shared data at different times, which can cause incorrect results, race conditions, or inconsistent data. Synchronization ensures processes execute in an organized and coordinated manner.

Task 3

Program for MPI_Reduce and MPI_Allreduce



Output



Difference between MPI_Reduce and MPI_Allreduce

- MPI_Reduce sends the final result only to the root process.
- MPI_Allreduce sends the final result to all processes.

So every process receives the result in MPI_Allreduce, while only the root process receives it in MPI_Reduce.

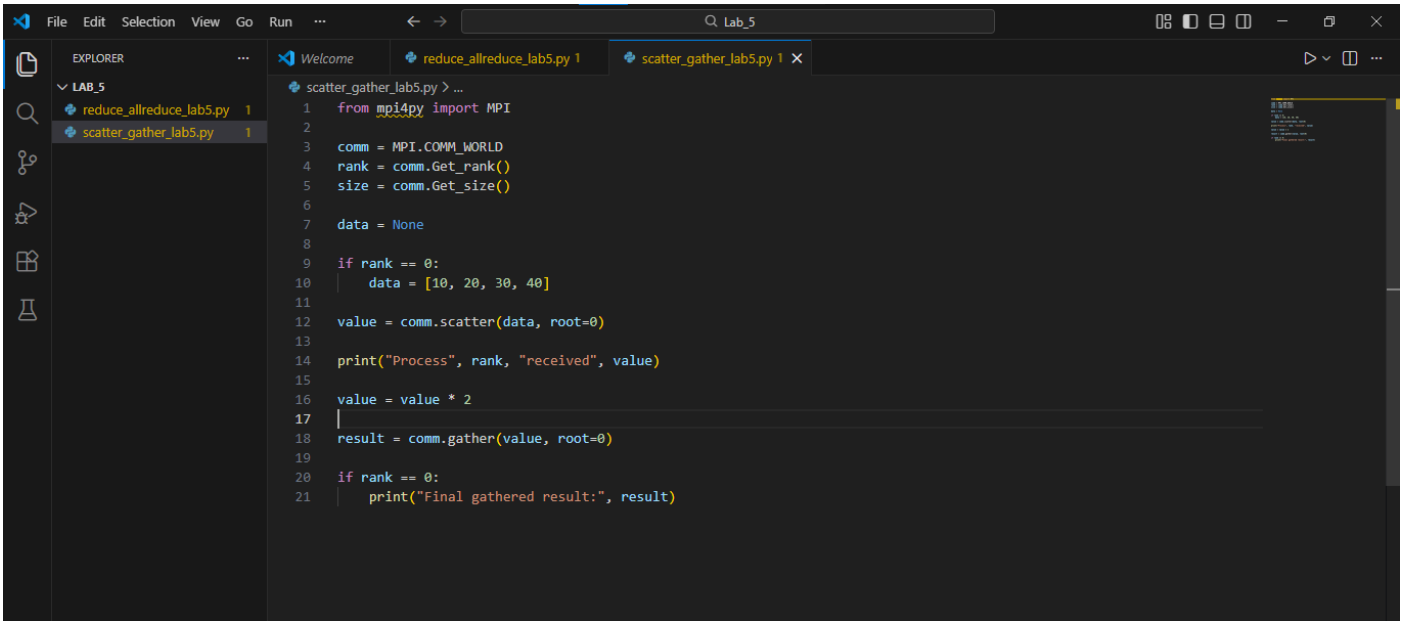
Task 4

What happens if MPI_Barrier is removed?

Answer:

If MPI_Barrier is removed, processes will not wait for each other. Some processes may execute faster and print output earlier, which can cause unordered output or incorrect synchronization. The program may still run, but execution will not be coordinated.

Task 5



Output

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_5>mpiexec -n 4 python scatter_gather_lab5.py
Process 1 received 20
Process 3 received 40
Process 2 received 30
Process 0 received 10
Final gathered result: [20, 40, 60, 80]

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_5>
```

Yes, MPI_Scatter and MPI_Gather can be used together in the same program. MPI_Scatter distributes data from the root process to all processes, while MPI_Gather collects processed data back to the root process.